RF Components: Characterization and Measurement (part -2)

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I. OBJECTIVE

(e) To measure amplifier gain, harmonics, the 1-dB compression point, and the third-order intercept point.

II. INTRODUCTION

An RF amplifier is an important active component in a radio-frequency receiver chain, where it is used to increase the power of weak received signals before further processing. Ideally, an amplifier should provide a constant gain over its operating frequency range while amplifying the input signal without introducing any distortion. In practice, however, the performance of an RF amplifier depends on both the frequency and the input signal power. At sufficiently high input powers, nonlinear effects become significant and can introduce harmonic components, gain compression, and intermodulation products.

The characterization of an RF amplifier therefore requires an understanding of both its amplification capability and its linearity. The gain of the amplifier determines the increase in signal power between its input and output and can vary with frequency. Measuring the gain over the operating frequency range provides information about the frequency response of the amplifier and the range over which it can provide useful amplification.

The linearity of the amplifier becomes particularly important when the input signal power is increased. A nonlinear amplifier can generate harmonics of a single input frequency, producing unwanted components at integer multiples of the fundamental frequency. As the input power is increased further, the amplifier eventually departs from its small-signal behaviour and its gain begins to decrease. The input power corresponding to a 1-dB reduction in gain from its small-signal value is known as the 1-dB compression point and provides an important measure of the amplifier's linear operating range.

When multiple RF signals are present simultaneously, nonlinear effects can also produce intermodulation products. For two closely spaced input frequencies f_1 and f_2, third-order nonlinearities generate components at

$$2f_1 - f_2 \tag{1}$$

and

$$2f_2 - f_1. \tag{2}$$

These third-order products are especially significant because they occur close to the original signals and can therefore interfere with desired signals. The relative growth of the fundamental and third-order intermodulation components with increasing input power provides a measure of amplifier linearity. Their extrapolated intersection defines the third-order intercept point (TOI), which is an important figure of merit for RF amplifier performance.

Thus, the characterization of the RF amplifier involves examining its frequency-dependent gain, the generation of harmonic distortion and gain compression at increasing input power, and its susceptibility to third-order intermodulation. Together, these measurements provide a comprehensive assessment of the amplifier's gain, usable operating range, and linearity.

III. THEORY

An RF amplifier is an active component used to increase the power of an RF signal. Amplifiers are essential elements of an RF receiver chain because the signals received by a radio telescope are generally very weak and require amplification before subsequent stages of signal processing. Ideally, an amplifier should increase the signal power without altering its frequency content or introducing distortion. In practice, however, an RF amplifier exhibits nonlinear behaviour when the input signal becomes sufficiently large.

The characterization of an RF amplifier therefore requires more than just measuring its gain. In this experiment, the amplifier is characterized in terms of

1. *small-signal gain,*
2. *harmonic generation,*
3. *the 1-dB compression point, and*
4. *the third-order intercept point (TOI).*

These quantities provide information about the amplification, frequency-dependent response, and linearity of the amplifier.

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A. Amplifier Gain

The power gain of an RF amplifier is defined as the ratio of the output power to the input power. In linear units,

$$G_{\text{linear}} = \frac{P_{\text{out}}}{P_{\text{in}}}. \tag{3}$$

Expressing the gain in decibels gives

$$G_{\text{dB}} = 10\log_{10}\left(\frac{P_{\text{out}}}{P_{\text{in}}}\right). \tag{4}$$

Since the input and output powers in the experiment are measured in dBm, the gain can conveniently be obtained by subtraction:

$$\boxed{G = P_{\text{out}} - P_{\text{in}}}. \tag{5}$$

Here, P_{in} is the amplifier input power in dBm, P_{out} is the amplifier output power in dBm, and G is the amplifier gain in dB.

For an amplifier operating in its small-signal region, the gain is approximately constant. If the small-signal gain is denoted by G_{SS}, then

$$P_{\text{out}} \approx P_{\text{in}} + G_{\text{SS}}. \tag{6}$$

Thus, an increase in input power by 1 dB produces approximately a 1 dB increase in output power as long as the amplifier remains in its linear region.

The amplifier gain can also depend on frequency. Therefore, measuring the input and output powers over the operating frequency range gives

$$\boxed{G(f) = P_{\text{out}}(f) - P_{\text{in}}(f)}. \tag{7}$$

A plot of gain versus frequency therefore provides information about the amplifier's frequency response and operating band.

B. Nonlinear Behaviour of an RF Amplifier

An ideal linear amplifier can be represented by

$$y(t) = a_1 x(t), \tag{8}$$

where $x(t)$ is the input signal, $y(t)$ is the output signal, and a_1 represents the linear amplification coefficient.

A practical RF amplifier can exhibit nonlinear behaviour and may be represented by a power-series expansion:

$$y(t) = a_1 x(t) + a_2 x^2(t) + a_3 x^3(t) + \cdots. \tag{9}$$

The first-order term represents the desired linear amplification, while the higher-order terms describe nonlinear distortion.

The second-order term,

$$a_2 x^2(t), \tag{10}$$

is responsible for second-order nonlinear products, whereas the third-order term,

$$a_3 x^3(t), \tag{11}$$

produces third-order distortion and third-order intermodulation products.

To understand harmonic generation, consider a single sinusoidal input,

$$x(t) = A\cos(\omega t), \tag{12}$$

where A is the input amplitude and

$$\omega = 2\pi f \tag{13}$$

is the angular frequency.

Substituting this into the nonlinear amplifier equation gives

$$y(t) = a_1 A\cos(\omega t) + a_2 A^2\cos^2(\omega t) + a_3 A^3\cos^3(\omega t) + \cdots. \tag{14}$$

Using the identities

$$\cos^2(\omega t) = \frac{1+\cos(2\omega t)}{2}, \tag{15}$$

and

$$\cos^3(\omega t) = \frac{3\cos(\omega t) + \cos(3\omega t)}{4}, \tag{16}$$

the output becomes

$$y(t) = a_1 A\cos(\omega t) + \frac{a_2 A^2}{2}\left[1+\cos(2\omega t)\right] + \frac{a_3 A^3}{4}\left[3\cos(\omega t) + \cos(3\omega t)\right] + \cdots. \tag{17}$$

Rearranging,

$$y(t) = \frac{a_2 A^2}{2} + \left(a_1 A + \frac{3a_3 A^3}{4}\right)\cos(\omega t) + \frac{a_2 A^2}{2}\cos(2\omega t) + \frac{a_3 A^3}{4}\cos(3\omega t) + \cdots. \tag{18}$$

This expression shows that the output consists of several frequency components. These include

$$f, \qquad 2f, \qquad 3f, \ldots \tag{19}$$

in addition to a DC component.

The component at f is the fundamental component, while the components at $2f$, $3f$, and higher multiples are the second, third, and higher harmonics, respectively.

For a perfectly linear amplifier, the higher-order coefficients would be absent and only the fundamental frequency would appear at the output. Therefore, the observation of harmonics in the output spectrum is a direct indication of nonlinear behaviour.

C. Harmonic Generation

When a single-frequency signal at frequency f is applied to a nonlinear amplifier, the output can contain harmonics at integer multiples of the input frequency:

$$f, \quad 2f, \quad 3f, \quad 4f, \ldots. \tag{20}$$

The second harmonic occurs at

$$f_2 = 2f, \tag{21}$$

while the third harmonic occurs at

$$f_3 = 3f. \tag{22}$$

The nonlinear terms determine the strength of these harmonic components. In particular, the second-order coefficient contributes to the second harmonic, while the third-order coefficient contributes to the third harmonic.

As the input power is increased, nonlinear effects become stronger. Consequently, the harmonic components generally become more significant relative to the fundamental.

The spectrum analyser can therefore be used to observe the fundamental and harmonic components and measure their corresponding power levels.

D. Small-Signal Operation and Gain Compression

At sufficiently low input powers, the amplifier operates in its small-signal region. In this region, the gain is approximately constant:

$$G \simeq G_{\mathrm{SS}}. \tag{23}$$

Consequently,

$$P_{\mathrm{out}} \simeq P_{\mathrm{in}} + G_{\mathrm{SS}}. \tag{24}$$

Therefore, on a plot of output power in dBm versus input power in dBm, the small-signal response has an approximate slope of

$$1 : 1. \tag{25}$$

As the input power is increased, the amplifier eventually approaches the limit of its available output swing. Nonlinear effects become increasingly important, and the output power begins to deviate from the ideal linear relationship.

The gain consequently starts to decrease with increasing input power. This phenomenon is known as gain compression.

The amplifier response can therefore be considered in three broad regions:

1. ***Linear or small-signal region:*** *the gain remains approximately constant.*
2. ***Compression region:*** *the gain gradually decreases as the input power is increased.*
3. ***Saturation region:*** *the output power approaches a maximum value and shows only a small increase with further increases in input power.*

E. 1-dB Compression Point

The 1-dB compression point is a standard measure of the maximum approximately linear operating level of an RF amplifier.

Let the small-signal gain be

$$G_{\mathrm{SS}}. \tag{26}$$

If the amplifier remained perfectly linear, the expected output power would be

$$P_{\mathrm{out,ideal}} = P_{\mathrm{in}} + G_{\mathrm{SS}}. \tag{27}$$

Because of compression, the actual output power becomes smaller than this ideal extrapolated value.

The 1-dB compression point is defined as the point at which the actual amplifier gain has decreased by 1 dB from its small-signal value:

$$\boxed{G_{\mathrm{1dB}} = G_{\mathrm{SS}} - 1 \text{ dB}}. \tag{28}$$

If $P_{\mathrm{in,1dB}}$ is the input power at the 1-dB compression point, then

$$G_{1\mathrm{dB}} = P_{\mathrm{out,1dB}} - P_{\mathrm{in,1dB}}. \tag{29}$$

Therefore,

$$\boxed{P_{\mathrm{out,1dB}} = P_{\mathrm{in,1dB}} + G_{\mathrm{SS}} - 1\ \mathrm{dB}}. \tag{30}$$

Thus, the 1-dB compression point is specified by the corresponding input power

$$\boxed{P_{\mathrm{in,1dB}}} \tag{31}$$

and output power

$$\boxed{P_{\mathrm{out,1dB}}}. \tag{32}$$

Experimentally, the gain is calculated at different input powers using

$$G(P_{\mathrm{in}}) = P_{\mathrm{out}} - P_{\mathrm{in}}. \tag{33}$$

The 1-dB compression point is reached when

$$G(P_{\mathrm{in}}) = G_{\mathrm{SS}} - 1\ \mathrm{dB}. \tag{34}$$

Hence, the 1-dB compression point provides a quantitative indication of the input-power level beyond which the amplifier can no longer maintain its small-signal gain.

F. Third-Order Intermodulation

When two sinusoidal signals are simultaneously applied to a nonlinear amplifier, additional frequency components are produced due to intermodulation.

Let the two input frequencies be f_1 and f_2. The input signal can be written as

$$x(t) = A_1 \cos(\omega_1 t) + A_2 \cos(\omega_2 t), \tag{35}$$

where

$$\omega_1 = 2\pi f_1, \qquad \omega_2 = 2\pi f_2. \tag{36}$$

The output contains the original fundamental frequencies

$$f_1, \qquad f_2, \tag{37}$$

as well as nonlinear products.

The third-order nonlinear terms generate, among other products, the two third-order intermodulation components closest to the fundamentals:

$$\boxed{f_{\mathrm{IM3,1}} = 2f_1 - f_2} \tag{38}$$

and

$$\boxed{f_{\mathrm{IM3,2}} = 2f_2 - f_1}. \tag{39}$$

These products are particularly important because they are located close to the original signals.

For closely spaced frequencies, let

$$f_2 = f_1 + \Delta f. \tag{40}$$

Then

$$2f_1 - f_2 = f_1 - \Delta f, \tag{41}$$

and

$$2f_2 - f_1 = f_2 + \Delta f. \tag{42}$$

Hence, the third-order intermodulation products appear immediately outside the two fundamental tones.

This makes third-order distortion especially significant in RF systems, where closely spaced signals may be present simultaneously.

G. Power Dependence of Fundamental and Third-Order Products

The fundamental output is primarily associated with the first-order response of the amplifier. Consequently, the fundamental output power increases approximately proportionally to the input power.

On a logarithmic power scale, this gives an approximate relationship

$$P_{\mathrm{out,fund}} \propto P_{\mathrm{in}}, \tag{43}$$

corresponding to a slope of

$$\boxed{1:1}. \tag{44}$$

The third-order intermodulation products originate from the third-order nonlinear term. Their amplitude is proportional to the cube of the input amplitude:

$$A_{\mathrm{IM3}} \propto A_{\mathrm{in}}^3. \tag{45}$$

Therefore, on the logarithmic power scale, the third-order intermodulation output increases much more rapidly with input power, with an approximate slope of

$$\boxed{3:1}. \tag{46}$$

Thus, the fundamental and third-order products follow approximately

$$P_{\text{fund}} \sim P_{\text{in}}, \tag{47}$$
$$P_{\text{IM3}} \sim 3P_{\text{in}}. \tag{48}$$

This difference in slopes is the basis of the third-order intercept measurement.

H. Third-Order Intercept Point

The third-order intercept point (TOI) is a figure of merit used to characterize the third-order linearity of an RF amplifier.

The fundamental output follows an approximately $1:1$ *slope with input power, while the third-order intermodulation output follows an approximately* $3:1$ *slope.*

If the measured trends are extrapolated beyond the actual operating range, the two lines eventually intersect. This extrapolated intersection is defined as the third-order intercept point.

The fundamental trend can be represented as

$$P_{\text{fund}} = P_{\text{in}} + C_1, \tag{49}$$

where C_1 *is a constant determined by the amplifier gain.*

Similarly, the third-order intermodulation trend can be represented as

$$P_{\text{IM3}} = 3P_{\text{in}} + C_3, \tag{50}$$

where C_3 *is another constant.*

At the theoretical intercept,

$$P_{\text{fund}} = P_{\text{IM3}}. \tag{51}$$

Therefore,

$$P_{\text{in}} + C_1 = 3P_{\text{in}} + C_3. \tag{52}$$

The input power corresponding to this extrapolated intersection is the input third-order intercept point, IIP3. The corresponding output power is the output third-order intercept point, OIP3.

The relation between the two is approximately

$$\boxed{\text{OIP3} = \text{IIP3} + G_{\text{SS}}}. \tag{53}$$

The TOI is a theoretical extrapolation rather than an operating point that is normally reached experimentally. In practice, the amplifier enters compression and eventually saturation before the fundamental and third-order curves actually intersect.

Therefore, a higher TOI generally indicates better third-order linearity because the third-order distortion remains relatively small over a larger input-power range.

IV. COMPONENTS REQUIRED

- *RF signal generator(s)*
- *Spectrum analyser*
- *Digital storage oscilloscope*
- *Power splitter/combiner*
- *RF amplifier and the required DC power supply*
- *Open-ended coaxial stub*
- *Short reference coaxial cable*
- *50 Ω terminations*
- *Suitable RF cables, adapters, and attenuators*

V. EXPERIMENTAL SETUP

The experimental setup for RF amplifier characterization is based on the transmission measurement arrangement shown as Setup A in the laboratory manual. The RF signal generator is used as the source of the input signal, the RF amplifier is connected as the device under test, and the output of the amplifier is monitored using a spectrum analyser.

For the amplifier measurements, the RF signal generator is connected to the input of the amplifier using suitable RF cables, adapters, and attenuators where required. The amplifier is powered using the specified external DC power supply. The output of the amplifier is then connected to the spectrum analyser, which measures the output power and allows the frequency components present in the output spectrum to be observed.

The general signal path can be represented as

$$\boxed{\textit{RF Signal Generator} \longrightarrow \textit{RF Amplifier} \longrightarrow \textit{Spectrum Analyser}} \tag{54}$$

For the frequency-dependent gain measurement, the input power is maintained at a safe fixed level while the frequency is varied over the operating range of the amplifier. The amplifier input power and corresponding output power are measured, allowing the gain to be calculated as

$$G = P_{\text{out}} - P_{\text{in}}. \tag{55}$$

For the single-tone nonlinear characterization, the signal generator is set to a selected frequency within the amplifier operating band. The input power is gradually increased in safe steps, while the spectrum analyser is used to record the fundamental output power and the visible harmonic components. The amplifier gain is calculated at each input-power level, allowing the onset of gain compression and the 1*-dB compression point to be determined.*

For the two-tone characterization, two RF signal generators are used to produce closely spaced input frequencies f_1 and f_2. Their outputs are combined using an RF power combiner before being applied to the amplifier. The amplifier output is connected to the spectrum analyser. The spectrum analyser is used to identify the fundamental components at f_1 and f_2 and the third-order intermodulation products at

$$2f_1 - f_2 \tag{56}$$

and

$$2f_2 - f_1. \tag{57}$$

For the two-tone amplifier characterization, two RF signal generators are used to generate two closely spaced frequencies f_1 and f_2. The outputs of the two signal generators are connected to the two input ports of an RF power combiner. The combined signal is then applied to the input of the RF amplifier, and the amplifier output is connected to the spectrum analyser. The spectrum analyser is used to observe the fundamental components at f_1 and f_2 and the third-order intermodulation products at $2f_1 - f_2$ and $2f_2 - f_1$.

The signal flow can be represented as

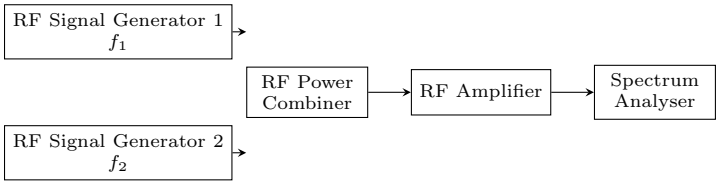


FIG. 1: Experimental setup for two-tone RF amplifier characterization.

with RF Signal Generator 2 supplying the second input frequency f_2 to the other input port of the RF power combiner. with the second signal generator supplying the second input frequency

The amplifier must be powered only after its supply voltage and polarity have been verified. The generator power should be kept within the safe operating range of the amplifier, and an external attenuator should be used if the expected amplifier output could exceed the maximum permitted input power of the spectrum analyser. The two signal-generator outputs must be combined using the RF power combiner rather than being connected directly together.

VI. OBSERVATIONS

The observations for the RF amplifier characterization were recorded for its frequency response, nonlinear response to increasing input power, and third-order intermodulation behaviour. The measured input and output powers were expressed in dBm, while the amplifier gain was calculated in dB using

$$G = P_{\text{out}} - P_{\text{in}}. \tag{58}$$

The observations corresponding to the amplifier characterization are presented below.

A. Gain versus Frequency

The amplifier input power was maintained at a constant value of $P_{\text{in}} = -10$ dBm while the frequency was varied over the operating range. The corresponding output power and gain were recorded.

TABLE I: Measured input power, output power, and gain as a function of frequency.

Frequency (MHz)	P_{in} **(dBm)**	P_{out} **(dBm)**	**Gain (dB)**
50	-10	13.81	23.81
150	-10	13.28	23.28
250	-10	12.92	22.92
350	-10	12.63	22.63
450	-10	12.52	22.52
550	-10	12.22	22.22
650	-10	12.23	22.23
750	-10	12.18	22.18
850	-10	11.87	21.87
950	-10	11.57	21.57
1050	-10	11.63	21.63
1150	-10	11.62	21.62
1250	-10	11.34	21.34
1350	-10	11.15	21.15
1450	-10	10.92	20.92
1550	-10	10.45	20.45
1650	-10	10.26	20.26
1750	-10	10.10	20.10
1850	-10	9.52	19.52
1950	-10	8.40	18.40
2000	-10	8.20	18.20

B. Harmonics and 1-dB Compression Point

For the nonlinear characterization, the input power was increased from -20 dBm to -3 dBm at a fixed frequency. The corresponding output power and gain were recorded. The visible harmonic power was also recorded where measurable.

TABLE II: Measured input power, output power, gain, and harmonic power during amplifier nonlinear characterization.

P_{in} (dBm)	P_{out} (dBm)	Gain (dB)	Harmonic Power (dBm)
-20	3.80	23.80	-44.64
-19	4.80	23.80	—
-18	5.81	23.81	—
-17	6.80	23.80	—
-16	7.78	23.78	—
-15	8.76	23.76	—
-14	9.73	23.73	—
-13	10.74	23.74	—
-12	11.70	23.70	—
-11	12.69	23.69	—
-10	13.62	23.62	—
-9	14.54	23.54	—
-8	15.47	23.47	—
-7	16.25	23.25	—
-6	16.85	22.85	—
-5	17.29	22.29	—
-4	17.70	21.70	—
-3	17.98	20.98	—

The small-signal gain was determined from the approximately constant gain region at low input powers. The gain begins to decrease as the input power is increased, indicating the onset of gain compression.

The small-signal gain is approximately

$$G_{\text{SS}} \approx 23.80 \text{ dB}. \tag{59}$$

Therefore, the gain corresponding to the 1*-dB compression point is*

$$G_{\text{1dB}} = G_{\text{SS}} - 1 \approx 22.80 \text{ dB}. \tag{60}$$

From interpolation of the measured gain data, the corresponding input and output powers are approximately

$$P_{\text{in,1dB}} \approx -5.91 \text{ dBm}, \tag{61}$$

and

$$P_{\text{out,1dB}} \approx 16.89 \text{ dBm}. \tag{62}$$

C. Third-Order Intercept

For the two-tone measurement, two closely spaced input frequencies were applied to the amplifier. The input frequencies were

$$f_1 = 13 \text{ MHz}, \qquad f_2 = 10 \text{ MHz}, \tag{63}$$

with an input power of 10 dBm *specified for the first signal generator. The fundamental output power and the power of an adjacent third-order intermodulation product were recorded for different input powers per tone.*

The relevant third-order intermodulation frequencies are

$$2f_1 - f_2 = 2(13) - 10 = 16 \text{ MHz}, \tag{64}$$

and

$$2f_2 - f_1 = 2(10) - 13 = 7 \text{ MHz}. \tag{65}$$

The measured observations are given in Table III.

TABLE III: Measured fundamental and third-order intermodulation powers.

P_{in} per tone (dBm)	P_{fund} (dBm)	P_{IM3} (dBm)
-10	10.36	-25.80
-9	11.36	-21.00
-8	12.24	-18.20
-7	13.20	-15.21
-6	14.00	-12.60
-5	14.80	-10.00

The fundamental output power increases approximately linearly with input power, whereas the third-order intermodulation product increases more rapidly. Linear fitting of the measured data gives approximately

$$\textit{Fundamental slope} \approx 0.888 : 1, \tag{66}$$

and

$$\textit{Third-order slope} \approx 3.063 : 1. \tag{67}$$

The extrapolated intersection of the fundamental and third-order trends gives the third-order intercept point. From the measured data, the extrapolated input and output intercept powers are approximately

$$\text{IIP3} \approx 6.20 \text{ dBm}, \tag{68}$$

and

$$\text{OIP3} \approx 24.83 \text{ dBm}. \tag{69}$$

These values represent extrapolated intercept points; the amplifier would normally enter compression before the actual intersection is reached.

VII. GRAPHS AND ANALYSIS

The measured data were analysed to characterize the RF amplifier in terms of its frequency response, gain compression, and third-order intermodulation behaviour. The gain was calculated from

$$G = P_{\text{out}} - P_{\text{in}}, \tag{70}$$

where P_{in} and P_{out} are expressed in dBm.

A. Gain versus Frequency

The amplifier gain was measured over the frequency range from 50 MHz *to* 2000 MHz *while maintaining the input power at approximately* −10 dBm. *The gain at each frequency was calculated using*

$$G(f) = P_{\text{out}}(f) - P_{\text{in}}(f). \tag{71}$$

The resulting variation of gain with frequency is shown in Fig. 2.

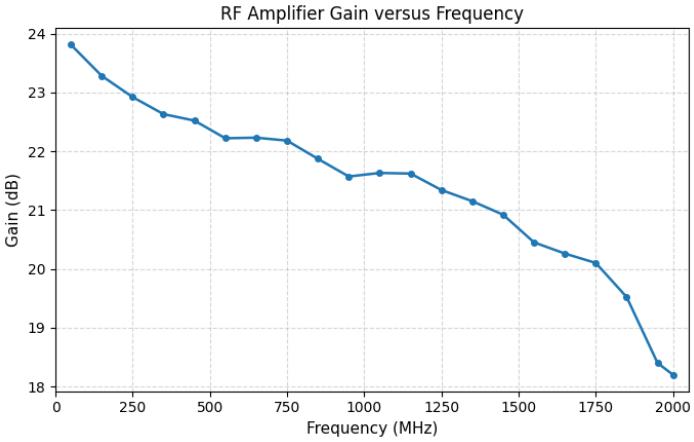


FIG. 2: Measured RF amplifier gain as a function of frequency.

From the graph, the amplifier exhibits its highest gain at the lower end of the measured frequency range. The gain is approximately 23.8 dB *at* 50 MHz *and decreases progressively with increasing frequency. At* 1000 MHz, *the gain is approximately* 21.6 dB, *while at* 2000 MHz *it falls to approximately* 18.2 dB.

Thus, the measured amplifier shows a clear decrease in gain at higher frequencies. The total decrease in gain over the measured frequency range is approximately

$$\Delta G = 23.8 - 18.2 = 5.6 \text{ dB}. \tag{72}$$

Therefore, within the measured range, the amplifier provides its largest gain at low frequencies and progressively lower gain towards 2 GHz.

B. Output Power versus Input Power

The output power was measured while increasing the amplifier input power from −20 dBm *to approximately* −3 dBm. *The measured output power is plotted together with the extrapolated small-signal response in Fig. 3.*

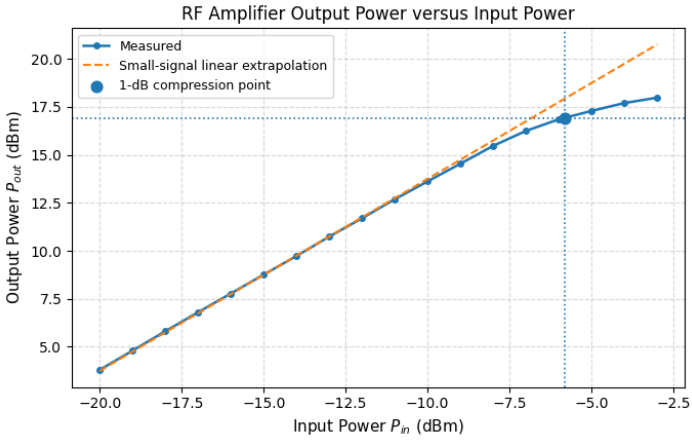


FIG. 3: Measured output power as a function of input power and extrapolated small-signal response.

At low input powers, the measured output power follows the small-signal linear extrapolation closely. This indicates that the amplifier operates approximately linearly in this region. As the input power is increased, the measured curve begins to deviate from the ideal linear response, indicating the onset of gain compression.

The small-signal gain obtained from the low-power region is

$$\boxed{G_{\text{SS}} = 23.75 \text{ dB}}. \tag{73}$$

The gain corresponding to the 1*-dB compression point is therefore*

$$G_{1\text{dB}} = G_{\text{SS}} - 1 \tag{74}$$

$$= 23.75 - 1 \tag{75}$$

$$= \boxed{22.75 \text{ dB}}. \tag{76}$$

From the intersection of the measured gain response with this 1*-dB-compressed gain level, the input power at the compression point is*

$$\boxed{P_{\text{in,1dB}} = -5.82 \text{ dBm}}. \tag{77}$$

The corresponding output power is

$$\boxed{P_{\text{out,1dB}} = 16.93 \text{ dBm}}. \tag{78}$$

Hence, the amplifier remains approximately linear up to an input power of about −5.82 dBm, *after which appreciable gain compression is observed.*

C. Gain versus Input Power

The gain was calculated for every measured input-power level using

$$G = P_{\text{out}} - P_{\text{in}}. \tag{79}$$

The resulting gain variation is shown in Fig. 4.

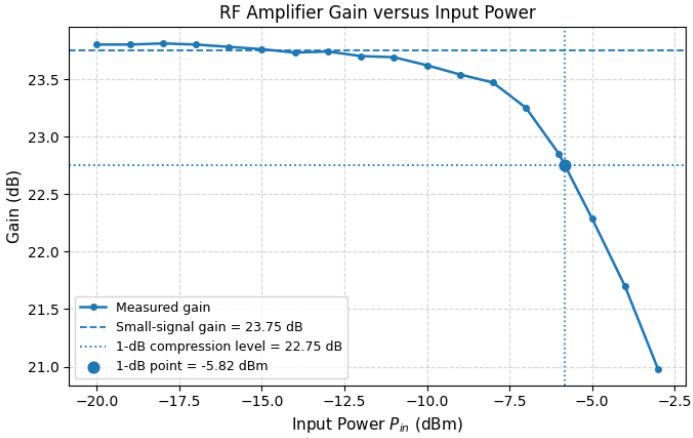


FIG. 4: RF amplifier gain as a function of input power.

At low input powers, the gain remains nearly constant at approximately 23.75 dB. *This region corresponds to the small-signal operating regime. As the input power is increased beyond approximately* −10 dBm, *the gain starts to decrease gradually.*

At

$$P_{\text{in}} = -5.82 \text{ dBm}, \tag{80}$$

the gain becomes

$$G = 22.75 \text{ dB}, \tag{81}$$

which is 1 dB *below the small-signal gain. This identifies the* 1*-dB compression point.*

The rapid decrease in gain at higher input powers demonstrates the nonlinear behaviour of the amplifier as it approaches its maximum linear output capability.

D. Third-Order Intercept Point

For the two-tone measurement, the fundamental output power and the third-order intermodulation product were measured for different input powers per tone. The measured data are plotted in Fig. 5.

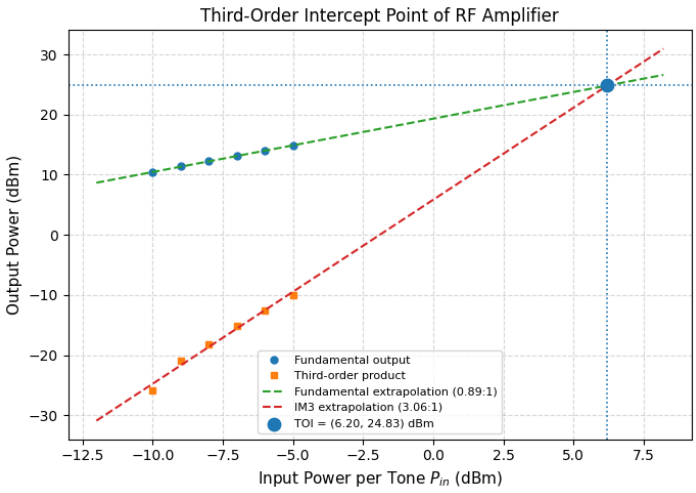


FIG. 5: Fundamental and third-order intermodulation output powers as a function of input power per tone.

The fundamental output power increases approximately linearly with input power, whereas the third-order intermodulation product increases much more rapidly. Linear fits to the measured data give the slopes

$$m_{\text{fund}} = 0.888, \tag{82}$$

and

$$m_{\text{IM3}} = 3.063. \tag{83}$$

These values are close to the theoretically expected 1 : 1 *and* 3 : 1 *slopes for the fundamental and third-order intermodulation components, respectively.*

The fitted fundamental and third-order responses can be written as

$$P_{\text{fund}} = 0.888 P_{\text{in}} + b_1, \tag{84}$$

and

$$P_{\text{IM3}} = 3.063 P_{\text{in}} + b_2. \tag{85}$$

The third-order intercept point is obtained by extrapolating these two fitted lines until they intersect:

$$P_{\text{fund}} = P_{\text{IM3}}. \tag{86}$$

Solving the two fitted equations gives the input third-order intercept point

$$\boxed{\text{IIP3} = 6.20 \text{ dBm}}. \tag{87}$$

The corresponding output power at the intersection is

$$\boxed{\text{OIP3} = 24.83 \text{ dBm}}. \tag{88}$$

Thus, the experimentally determined third-order intercept point is

$$\boxed{(\text{IIP3}, \text{OIP3}) = (6.20,\ 24.83) \text{ dBm}.} \tag{89}$$

It should be noted that the intercept point is obtained by extrapolation. The amplifier reaches compression before the fundamental and third-order curves physically intersect during normal operation.

E. Additional Experimental Observations

Some additional observations were recorded during the experiment regarding the effect of the RF power combiner and the RF cable losses.

At a low input power of approximately -20 dBm*, the amplifier was observed to provide a gain of approximately* 23 dB*, which is consistent with the small-signal gain obtained from the measured data. When the input power was increased towards* -3 dBm*, the observed gain reduced to approximately* 20 dB*, indicating the onset of significant gain compression. The measured spreadsheet value at* -3 dBm *is approximately* 20.98 dB*, so the handwritten observation is consistent with the measured trend.*

The following additional points were noted during the two-tone measurement:

- *At low input power, approximately* -20 dBm*, the effect of the power combiner on the measured amplifier response was observed to be negligible within the experimental measurement.*
- *For the two-tone arrangement, each signal generator output undergoes approximately* 3 dB *of ideal combining loss. Thus, the two signals combine at the output of the power combiner rather than being connected directly to each other.*
- *An overall cable/wiring loss of approximately* 1.63 dB *was accounted for in the experimental setup.*
- *The measured cable and combiner losses should be considered when comparing the nominal generator power with the actual power reaching the amplifier input.*

The handwritten experimental notes therefore provide an additional practical interpretation of the measured results: at low power the amplifier operates close to its small-signal gain, whereas increasing the input power causes the gain to decrease due to compression.

F. Summary of Results

The principal results obtained from the amplifier characterization are summarized in Table IV.

TABLE IV: Summary of RF amplifier characterization results.

Parameter	Measured Value
Maximum measured gain	23.8 dB
Gain at 1000 MHz	21.6 dB
Gain at 2000 MHz	18.2 dB
Small-signal gain, G_{SS}	23.75 dB
1-dB compression gain	22.75 dB
$P_{\text{in,1dB}}$	-5.82 dBm
$P_{\text{out,1dB}}$	16.93 dBm
Fundamental fitted slope	0.888 : 1
Third-order fitted slope	3.063 : 1
IIP3	6.20 dBm
OIP3	24.83 dBm
Overall cable/wiring loss	1.63 dB

VIII. CALCULATIONS

The measured quantities were used to calculate the amplifier gain, 1*-dB compression point, and third-order intercept point. The calculations are presented below.*

A. Calculation of Amplifier Gain

The amplifier gain in decibels is calculated from the difference between the output and input powers:

$$G = P_{\text{out}} - P_{\text{in}}. \tag{90}$$

For example, at an input power of

$$P_{\text{in}} = -10 \text{ dBm}, \tag{91}$$

and an output power of

$$P_{\text{out}} = 13.75 \text{ dBm}, \tag{92}$$

the gain is

$$G = 13.75 - (-10) \tag{93}$$

$$= 23.75 \text{ dB}. \tag{94}$$

Thus,

$$\boxed{G = 23.75 \text{ dB}}. \tag{95}$$

The same relation was used for all the measured input and output power values.

B. Gain Variation with Frequency

At 50 MHz*, the measured gain was approximately*

$$G_{50} \approx 23.8 \text{ dB}. \tag{96}$$

At 2000 MHz*, the measured gain was approximately*

$$G_{2000} \approx 18.2 \text{ dB}. \tag{97}$$

Hence, the change in gain over the measured frequency range is

$$\begin{aligned} \Delta G &= G_{50} - G_{2000} && (98)\\ &= 23.8 - 18.2 && (99)\\ &= 5.6 \text{ dB}. && (100) \end{aligned}$$

Therefore, the amplifier gain decreases by approximately

$$\boxed{\Delta G \approx 5.6 \text{ dB}} \tag{101}$$

between 50 MHz *and* 2000 MHz*.*

C. Calculation of the 1-dB Compression Point

The small-signal gain was obtained from the approximately constant gain region at low input powers. The measured value is

$$\boxed{G_{\text{SS}} = 23.75 \text{ dB}}. \tag{102}$$

The gain at the 1*-dB compression point is therefore*

$$\begin{aligned} G_{1\text{dB}} &= G_{\text{SS}} - 1 && (103)\\ &= 23.75 - 1 && (104)\\ &= 22.75 \text{ dB}. && (105) \end{aligned}$$

Thus,

$$\boxed{G_{1\text{dB}} = 22.75 \text{ dB}}. \tag{106}$$

From the gain-versus-input-power graph, the measured gain reaches 22.75 dB *at an input power of approximately*

$$\boxed{P_{\text{in,1dB}} = -5.82 \text{ dBm}}. \tag{107}$$

The corresponding output power obtained from the P_{out} *versus* P_{in} *graph is*

$$\boxed{P_{\text{out,1dB}} = 16.93 \text{ dBm}}. \tag{108}$$

The result can also be verified directly from the definition of gain:

$$\begin{aligned} G_{1\text{dB}} &= P_{\text{out,1dB}} - P_{\text{in,1dB}} && (109)\\ &= 16.93 - (-5.82) && (110)\\ &= 22.75 \text{ dB}. && (111) \end{aligned}$$

Hence, the calculated value exactly corresponds to the required 1*-dB reduction from the small-signal gain.*

D. Calculation of Third-Order Intercept Point

For the two-tone measurement, the fundamental and third-order intermodulation output powers were plotted against the input power per tone. Linear fits to the measured data gave the slopes

$$m_{\text{fund}} = 0.888 \tag{112}$$

and

$$m_{\text{IM3}} = 3.063. \tag{113}$$

The fitted equations can be written in the general form

$$P_{\text{fund}} = m_{\text{fund}} P_{\text{in}} + b_{\text{fund}}, \tag{114}$$

and

$$P_{\text{IM3}} = m_{\text{IM3}} P_{\text{in}} + b_{\text{IM3}}. \tag{115}$$

The third-order intercept occurs at the extrapolated intersection of these two lines:

$$P_{\text{fund}} = P_{\text{IM3}}. \tag{116}$$

Therefore,

$$m_{\text{fund}} P_{\text{in}} + b_{\text{fund}} = m_{\text{IM3}} P_{\text{in}} + b_{\text{IM3}}. \tag{117}$$

Rearranging,

$$P_{\text{in,TOI}} = \frac{b_{\text{IM3}} - b_{\text{fund}}}{m_{\text{fund}} - m_{\text{IM3}}}. \tag{118}$$

Using the fitted experimental lines gives

$$\boxed{P_{\text{in,TOI}} = \text{IIP3} = 6.20 \text{ dBm}}. \tag{119}$$

Substitution of this value into either fitted line gives the corresponding output intercept power:

$$\boxed{P_{\text{out,TOI}} = \text{OIP3} = 24.83 \text{ dBm}}. \tag{120}$$

Therefore,

$$\boxed{\text{IIP3} = 6.20 \text{ dBm}, \qquad \text{OIP3} = 24.83 \text{ dBm}.} \tag{121}$$

E. Comparison with Theoretical Slopes

For an ideal third-order nonlinear amplifier, the fundamental component is expected to increase with an approximately $1:1$ *slope, whereas the third-order intermodulation product is expected to increase with an approximately* $3:1$ *slope.*

The measured fundamental slope was

$$m_{\text{fund}} = 0.888. \tag{122}$$

Its percentage deviation from the ideal value of 1 *is*

$$\%\text{deviation}_{\text{fund}} = \left|\frac{0.888 - 1}{1}\right| \times 100 \tag{123}$$

$$= 11.2\%. \tag{124}$$

Similarly, the measured third-order slope was

$$m_{\text{IM3}} = 3.063. \tag{125}$$

Its percentage deviation from the ideal value of 3 *is*

$$\%\text{deviation}_{\text{IM3}} = \left|\frac{3.063 - 3}{3}\right| \times 100 \tag{126}$$

$$\approx 2.1\%. \tag{127}$$

Thus, the measured third-order slope is in close agreement with the expected $3:1$ *behaviour, while the fundamental slope shows a larger deviation from the ideal* $1:1$ *behaviour.*

IX. ERROR ANALYSIS

The experimental results are affected by several sources of uncertainty associated with the RF instruments, cables, connectors, power combiner, and the procedure used to determine the characteristic points. Since the supplied experimental data do not specify numerical uncertainties for the signal generator or spectrum analyser, a numerical instrument-based percentage error cannot be assigned reliably. The following analysis therefore considers the principal sources of experimental error and the deviations that can be quantified from the measured data.

A. Error in Gain Measurement

The gain is calculated using

$$G = P_{\text{out}} - P_{\text{in}}. \tag{128}$$

If the uncertainties in the input and output power measurements are δP_{in} *and* δP_{out}*, respectively, then the maximum possible uncertainty in the calculated gain is*

$$\boxed{\delta G = \delta P_{\text{out}} + \delta P_{\text{in}}}. \tag{129}$$

If independent random uncertainties are considered instead, the uncertainty may be combined in quadrature:

$$\boxed{\delta G = \sqrt{(\delta P_{\text{out}})^2 + (\delta P_{\text{in}})^2}.} \tag{130}$$

The actual numerical value of δG *cannot be determined from the available data because the measurement uncertainty/specification of the instruments was not provided.*

B. Error in the 1-dB Compression Point

The 1*-dB compression point is determined from the intersection of the measured gain curve with the level*

$$G_{\text{SS}} - 1. \tag{131}$$

Consequently, any uncertainty in the measured gain directly affects the location of the compression point.

In addition, the experimental gain data are discrete. Therefore, the exact compression point does not necessarily correspond to one of the measured input-power values and must be obtained by interpolation. For the present data, this procedure gives

$$P_{\text{in,1dB}} = -5.82 \text{ dBm}. \tag{132}$$

The uncertainty in this value depends on the spacing between the measured input-power points and the uncertainty in the corresponding gain measurements.

C. Error in the TOI Determination

The TOI is obtained by extrapolating two fitted lines. Therefore, it is particularly sensitive to deviations in the measured slopes and intercepts.

The ideal slopes are

$$m_{\text{fund,ideal}} = 1 \tag{133}$$

and

$$m_{\mathrm{IM3,ideal}} = 3. \tag{134}$$

The measured slopes were

$$m_{\mathrm{fund}} = 0.888 \tag{135}$$

and

$$m_{\mathrm{IM3}} = 3.063. \tag{136}$$

The corresponding deviations were found to be

$$\boxed{\%\mathrm{deviation}_{\mathrm{fund}} = 11.2\%} \tag{137}$$

and

$$\boxed{\%\mathrm{deviation}_{\mathrm{IM3}} \approx 2.1\%.} \tag{138}$$

The relatively small deviation of the IM3 slope indicates that the third-order intermodulation behaviour is well represented by the expected cubic dependence. The larger deviation of the fundamental slope may arise from experimental non-idealities and the limited number of measured points used for the linear fit.

Since the TOI is obtained by extrapolation, even small changes in the fitted slopes or intercepts can produce a noticeable change in the intersection point. Therefore, the calculated values

$$\mathrm{IIP3} = 6.20 \text{ dBm} \tag{139}$$

and

$$\mathrm{OIP3} = 24.83 \text{ dBm} \tag{140}$$

should be regarded as experimentally determined extrapolated values rather than directly measured operating points.

D. Cable and Connector Losses

An overall cable/wiring loss of approximately

$$\boxed{1.63 \text{ dB}} \tag{141}$$

was noted during the experiment. Cable, connector, and adapter losses can alter the actual RF power reaching the amplifier and the power delivered to the measuring instrument. If these losses are not completely accounted for, the measured gain and input-power values can differ from the intrinsic amplifier values.

The RF power combiner used for the two-tone measurement also introduces insertion loss. The handwritten experimental observation indicates an approximately 3 dB *combining loss for each signal path. Such losses must be considered when comparing the nominal generator settings with the actual power incident on the amplifier.*

Overall, the measured results show the expected physical behaviour of an RF amplifier: the gain decreases with frequency, the gain remains approximately constant at low input powers and subsequently compresses at higher powers, and the third-order intermodulation product increases substantially faster than the fundamental signal. The measured $3.063 : 1$ *IM3 slope is particularly close to the expected* $3 : 1$ *theoretical behaviour.*

X. RESULTS

The RF amplifier characterization yielded the following principal results:

- *The amplifier gain decreased from approximately* 23.8 dB *at* 50 MHz *to* 18.2 dB *at* 2000 MHz*.*
- *The small-signal gain was found to be*

$$G_{\mathrm{SS}} = 23.75 \text{ dB}.$$

- *The* 1*-dB compression point was obtained at*

$$P_{\mathrm{in,1dB}} = -5.82 \text{ dBm}, \qquad P_{\mathrm{out,1dB}} = 16.93 \text{ dBm}.$$

- *The measured slopes of the fundamental and third-order intermodulation products were* $0.888 : 1$ *and* $3.063 : 1$*, respectively, consistent with the expected* $1 : 1$ *and* $3 : 1$ *behaviour.*
- *The third-order intercept point was determined as*

$$\mathrm{IIP3} = 6.20 \text{ dBm}, \qquad \mathrm{OIP3} = 24.83 \text{ dBm}.$$

Overall, the amplifier exhibited decreasing gain with frequency and increasing nonlinear distortion with input power, with gain compression occurring at higher input levels.

XI. CONCLUSION

The RF amplifier characterization led to the following key findings and concepts:

1. *The amplifier gain was found to decrease with increasing frequency, falling from approximately* 23.8 dB *at* 50 MHz *to* 18.2 dB *at* 2000 MHz*.*
2. *The small-signal gain of the amplifier was determined to be* 23.75 dB*, representing the approximately linear operating region of the amplifier.*

3. *The* 1*-dB compression point was obtained at an input power of* -5.82 dBm*, demonstrating the onset of significant gain compression at higher input powers.*

4. *The generation of harmonic and intermodulation components demonstrated the nonlinear behaviour of a practical RF amplifier.*

5. *The measured fundamental and third-order intermodulation slopes were approximately* $0.888:1$ *and* $3.063:1$*, respectively, illustrating the characteristic* $1:1$ *and* $3:1$ *power dependence of the fundamental and third-order components.*

6. *The third-order intercept point was determined to be* IIP3 $= 6.20$ dBm *and* OIP3 $= 24.83$ dBm*.*

7. *The experiment demonstrated the importance of amplifier linearity, gain compression, harmonic distortion, and third-order intermodulation in RF system performance.*

8. *The core concepts learned include RF amplifier gain and frequency response, nonlinear amplification, harmonic generation,* 1*-dB compression, third-order intermodulation, and the use of TOI as a measure of amplifier linearity.*

XII. QUESTION ASKED IN VIVA

A. Observation of Mixer Output Using an Oscilloscope

A mixer performs frequency conversion by multiplying two input signals rather than simply adding them together. Let the RF and LO input signals be

$$v_{\mathrm{RF}}(t) = A\cos(2\pi f_{\mathrm{RF}}t) \tag{142}$$

and

$$v_{\mathrm{LO}}(t) = B\cos(2\pi f_{\mathrm{LO}}t). \tag{143}$$

The mixer output can ideally be represented as being proportional to the product of the two signals:

$$v_{\mathrm{out}}(t) \propto v_{\mathrm{RF}}(t)v_{\mathrm{LO}}(t). \tag{144}$$

Using the trigonometric identity

$$\cos A\cos B = \frac{1}{2}\left[\cos(A+B)+\cos(A-B)\right], \tag{145}$$

the output becomes

$$v_{\mathrm{out}}(t) \propto \frac{AB}{2}\left[\cos\left(2\pi(f_{\mathrm{RF}}+f_{\mathrm{LO}})t\right)+\cos\left(2\pi(f_{\mathrm{RF}}-f_{\mathrm{LO}})t\right)\right]. \tag{146}$$

Therefore, the mixer produces two main frequency components given by

$$f_{\mathrm{sum}} = f_{\mathrm{RF}} + f_{\mathrm{LO}} \tag{147}$$

and

$$f_{\mathrm{diff}} = \left|f_{\mathrm{RF}} - f_{\mathrm{LO}}\right|. \tag{148}$$

For example, if

$$f_{\mathrm{RF}} = 30\ \mathrm{MHz} \tag{149}$$

and

$$f_{\mathrm{LO}} = 10\ \mathrm{MHz}, \tag{150}$$

the mixer produces components at

$$f_{\mathrm{sum}} = 30 + 10 = 40\ \mathrm{MHz} \tag{151}$$

and

$$f_{\mathrm{diff}} = |30 - 10| = 20\ \mathrm{MHz}. \tag{152}$$

When the mixer output is connected to a spectrum analyser, these frequency components can be observed as separate peaks at 20 *MHz and* 40 *MHz. The spectrum analyser separates the output according to frequency, making the frequency-conversion process easy to identify.*

If the same mixer output is connected to an oscilloscope instead, the oscilloscope does not separate the individual frequency components. It displays the total voltage waveform as a function of time. For the above example, the waveform can be written as

$$v_{\mathrm{out}}(t) = V_1\cos(2\pi 40t) + V_2\cos(2\pi 20t). \tag{153}$$

For approximately equal amplitudes, this can be expressed using the sum-to- product identity as

$$\cos(2\pi 40t) + \cos(2\pi 20t) = 2\cos(2\pi 30t)\cos(2\pi 10t). \tag{154}$$

Thus, the oscilloscope shows a combined time-domain waveform in which a faster oscillation is modulated by a slower variation in amplitude. It does not directly show separate 20 *MHz and* 40 *MHz peaks.*

Therefore, the two instruments provide different views of the same mixer output:

Spectrum analyser:	*shows the individual frequency components;*
Oscilloscope:	*shows the combined voltage waveform in time.*

(155)

XIII. REFERENCES

1. Lab Mannual of SPA617
2. Wikipedia
3. Components and Measurements, YouTube video playlist: YouTube instructional video playlist.
4. AI tool for more conceptual understanding.

XIV. DECLARATION OF AI USAGE

I have used AI tools in the following ways for the completion of my lab report:

- *Formatting the theory content into various sub sections and concise writing of the same after I studied them from various sources. It helped me compile them well.*
- *Formatting of graph in python (lines along with labels) and the algorithm to find the cut off frequency from the graph.*

However ,the calculations, error analysis, data acquisition and processing, conclusions, key takeaways, results were all self written from scratch and there is absolutely no AI aid taken over those sections.

XV. ACKNOWLEDGMENT

I would like to thank our course instructor Dr. Mugundhan V. , our REO - Jayasandhya Meenakshinathan and our TA - Banhiman Ghosh, without their help this lab experiment would have been very difficult to complete.